

JULIEN SLUK

Backend-Focused Software Developer | Automation Engineer | Lead Trainer

Singapore | [GitHub](#) | [Portfolio](#) | [LinkedIn](#)

Professional Summary

Backend-focused software developer and automation engineer with a Diploma in Information Technology from Ngee Ann Polytechnic, one year of automation experience at OCBC Bank, and current experience as Lead Trainer for Computer Science at Zenitant. Builds monitoring platforms, workflow automation systems, bots, dashboards, and practical tools that reduce manual work and improve reliability. Experienced with Python, FastAPI, Selenium, SQL, Lua, APIs, automation scripts, and technical training.

Technical Skills

- Programming: Python, JavaScript, C#, SQL, HTML, CSS, Lua
- Backend & APIs: FastAPI, REST APIs, API design, JSON, CRUD workflows
- Automation: Selenium, Excel VBA, Google Apps Script, UiPath, Telegram Bots, CC:Tweaked turtle automation
- Data / ML: scikit-learn, OpenCV, clustering, prediction models, Chart.js, Leaflet
- Tools / Systems: Git, Docker, VS Code, Visual Studio, Power BI, SQLite, MongoDB, AWS, Azure, Windows, Linux, SAP HR systems exposure

Experience

Zenitant - Lead Trainer for Computer Science | 2026-Present

- Deliver computer science and technology training, guiding learners through programming, computational thinking, web development, automation, AI tools, and project-based activities.
- Translate technical concepts into beginner-friendly lessons, strengthening communication, leadership, and technical mentoring skills.

OCBC Bank - Automation Intern | 1 Year

- Built Python and Selenium automation for online banking service monitoring before 5AM, generating email reports with node-level service health and error diagnostics.
- Developed Python + RegEx file storage maintenance automation to delete targeted log files older than 90 days, reducing manual processing time by approximately 90%.
- Worked on CIF query performance testing, SSL certificate automation, database exporter tools, and database-to-Confluence workflows.

National Service - Automation Development & HR Systems Support | 2024-2026

- Worked with SAP-based HR systems and built internal automation tools for HR and operational workflows.
- Developed HR dashboard automation using Excel VBA, Google Calendar scheduling automation, internal chatbot tools, and Telegram Parade State Bot for attendance reporting.

Selected Projects

- PulseWatch Monitoring Platform: FastAPI monitoring platform with automated uptime checks, response-time history tracking, background workers, database persistence, and Chart.js dashboard visualization.
- Food Donation Platform: Full-stack donation/request platform with matching logic, machine learning demand prediction, and Leaflet map visualization.
- Minecraft CC:Tweaked Lua Auto Chest Sorter: Lua turtle automation system using x/y coordinate movement, double-chest layout rules, item pattern categorization, safe movement checks, and overflow fallback across a 16 double-chest wall.
- Trading Bot with Telegram Alerts: Python automation project integrated with Interactive Brokers API for strategy execution and Telegram trade notifications.

- Object Recognition Tool: Python/OpenCV computer vision project for detecting and classifying objects being held.

Education

- Singapore Institute of Technology - Computer Science, starting August 2026
- Ngee Ann Polytechnic - Diploma in Information Technology

Selected Polytechnic Projects

- Verified Aid Resource Website: Grade A project helping users find verified aid resources such as laptops and subsidies; included forums and feedback functionality.
- CRM Fashion Website: multi-role system for Sales, Customer, Marketing Manager, and Product Manager views.
- UiPath Financial Workflow: automated extraction of P/E and stock metrics to support business report generation.
- Machine Learning: HR promotion prediction using KPI/training metrics and Airbnb price prediction using location data.
- Analytics: K-means clustering, hierarchical clustering, text analysis, and association rule mining.
- Other projects: Building Placement Game, Movie Ticketing System, Anime Gacha Website, Cruise SQL, Credit Card SQL, Flight SQL, MongoDB database projects.

Freelance & Side Projects

- Library overdue email automation system sending 600+ emails and reducing approximately 20 hours of manual work.
- Teacher workload matrix multiplier for workload allocation across subjects and academic levels.
- Telegram-based inventory ordering system with Google Sheets backend.
- Bicycle GMT schedule auto-updater, auto car depreciation calculator, Capex/Opex calculator, and law case time-balancing payroll system.

Certifications

- IBM certifications
- Scrum certifications
- UiPath certifications
- Workato certifications
- NLP, Machine Learning, Data Fluency, Artificial Intelligence, and Neural Networks certifications

Leadership, Awards & Activities

- Team lead across Polytechnic projects, OCBC internship projects, and National Service technology initiatives.
- Outstanding CFC recognition for leading technology development initiatives.
- Received director recognition letters and appreciation emails for automation contributions.
- Student leader; Secondary School Badminton Team; Polytechnic event support committee.

Links

GitHub: <https://github.com/JulienSLuk>

Portfolio: <https://juliensluk.github.io>

LinkedIn: <https://www.linkedin.com/in/julien-setiawan-lukman-532034211>